

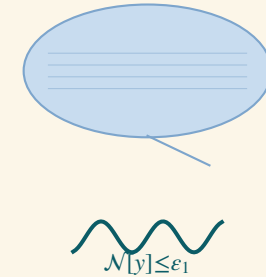
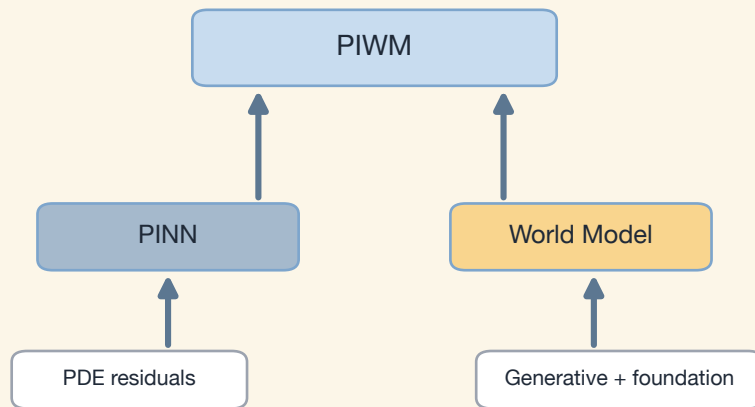
Physics-Informed World Models for Medical Imaging

A survey: foundations • integration mechanisms • applications

Sec 2 + Sec 5.1. Foundations & Definition

Classification by physical-constraint type

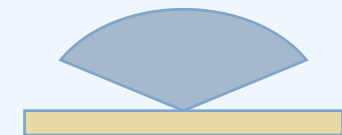
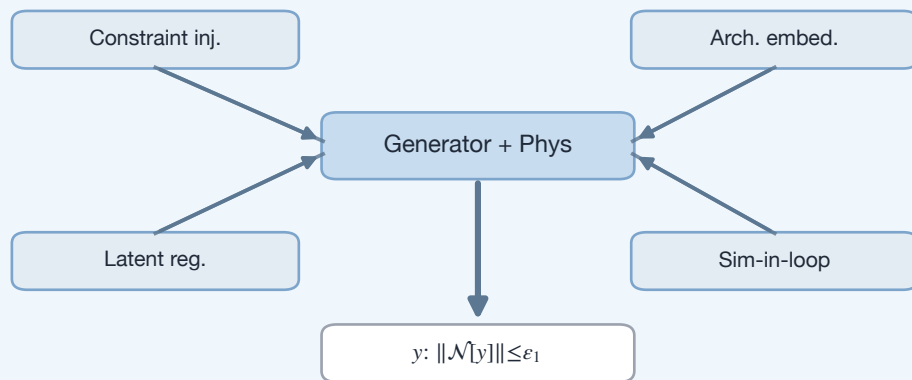
- ☐ 2.1 World Models: from theory to medical imaging
- ☐ 2.2 PINNs, physics-encoded architectures, neural operators
- ☐ 2.3 Medical imaging AI — current state & capability gaps
- ☐ 2.4 PIWM formal definition + physical-constraint-type axis
- ☐ 5.1 Dual-ceiling resolution & 4 integration mechanisms



Sec 5. PIWM Integration Mechanisms

Classification by integration strategy

- ☐ 5.1.2(a) Constraint injection — soft loss: $\mathcal{L}_{\text{WM}} + \lambda_{\text{phys}} \|\mathcal{N}[y]\|^2$
- ☐ 5.1.2(b) Architecture embedding — hard guarantee via $\mathcal{T}_{\text{phys}}$
- ☐ 5.1.2(c) Latent-space regulation — physics regulariser on z
- ☐ 5.1.2(d) Simulation-in-the-loop — train on $\mathcal{D}_{\text{real}} \cup \mathcal{D}_{\text{sim}}$
- ☐ 5.3 Four emergent capabilities (resolve world-model ceilings 1:1)

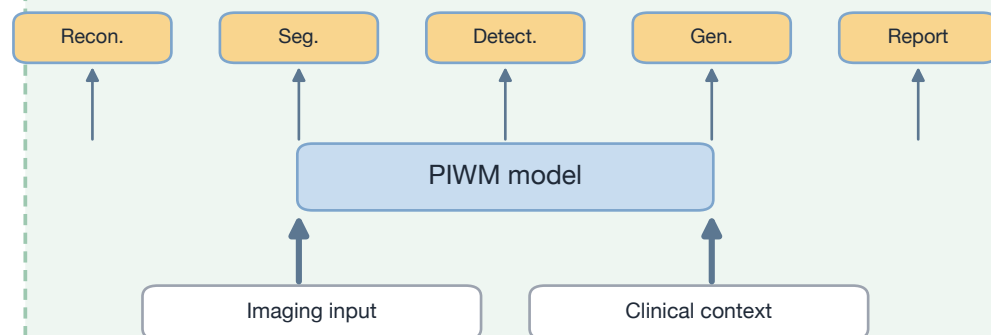


4 mechanisms
× 5 tasks

Sec 5.2 + Sec 8. Applications & Roadmap

Five medical-imaging tasks across radiology + pathology

- ☐ 5.2.1 Reconstruction — diffusion + data-consistency layers
- ☐ 5.2.2 Segmentation — topology-aware + biomechanically informed
- ☐ 5.2.3 Detection — physics-plausibility filters
- ☐ 5.2.4 Generation — Bloch-consistent synthesis + counterfactuals
- ☐ 8.2 Four-phase roadmap toward clinically trustworthy AI (Phase 4)



Radiology + Pathology